Can Beslendi[1]*, Michael Fujarski[1], Sarah Sandmann[1], Julian Varghese[1]
[1]Institute of Medical Informatics, University of Münster, Münster, Germany
*Mail: canbeslendi@live.de

Institut für Medizinische Informatik

Exploring Gene-Disease Associations by Gene Network Analysis on PubMed

Introduction

Assembling valid and thorough information on genes and their relation in different disease entities is a challenging task. For example, searching for a gene like "TP53" in PubMed returns around 25,000 results. To extract the information of interest, tedious manual screening would be necessary.
To overcome these limitations, we developed a website, which facilitates state-of-the-art text mining architectures to find matching associations on PubMed. Results are efficiently and intuitively visualized by means of an interactive gene network. Associated publications can be accessed directly.

Methods

A BERT-based model[1], which relies on machine learning, was pre-trained on PubMed articles. For fine-tuning, we decided to use and modify the pre-processed GAD and EU-ADR dataset from the BioBERT-paper[2]. Since these datasets were created in a semi-automatic way[3] and are thus weakly labelled, we performed a manual re-annotation of the whole dataset, resulting in high-quality relation extraction datasets.

Results

The trained model can now effectively discriminate negations and semantic correlations. Examples in *Fig. 1*:

1. The **@GENE$** gene codon 72 polymorphism may play a role in susceptibility to **@DISEASE$.**
   LABEL: 1, score: 0.9999103546142578
2. Genetic variation in **@GENE$** is thus unlikely to have a major impact on susceptibility to **@DISEASE$.**
   LABEL: 0, score: 0.9998624324798584
3. We found out that **@GENE$**, but not COX-1, contributes to a higher risk for **@DISEASE$.**
   LABEL: 1, score: 0.9999090433120728
4. We found out that COX-2, but not **@GENE$**, contributes to a higher risk for **@DISEASE$.**
   LABEL: 0, score: 0.9729235172271729

Fig. 1: The model accepts a string as an input, where the entities of interest must have the masks *@GENE$* or *@DISEASE$*. After processing, the classifier outputs a label (0: no correlation, 1: correlation between the masks) and a confidence-score.

Note in *Fig. 1* how in the same sentences 3 and 4, but with different masks, the model can distinguish between different gene mentions and its meaning in context.
This model provides the backend for the website with its user-interface in *Fig. 2* and *3*

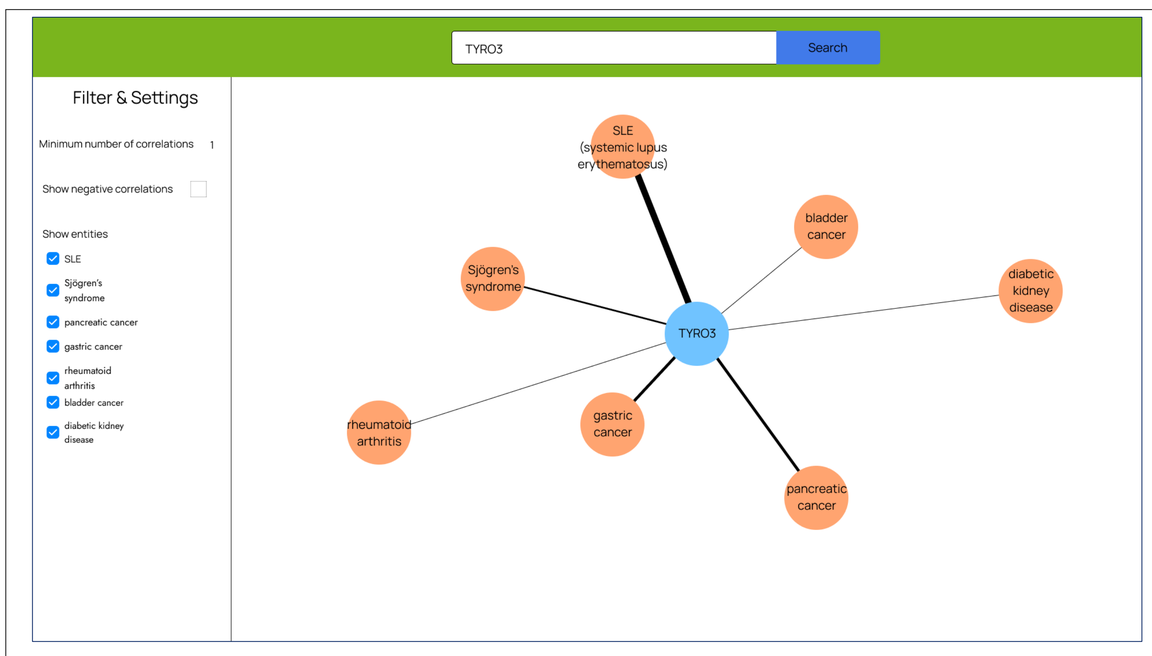

Fig. 2: After entering a gene in the search bar, nodes of other diseases are shown, which were classified as correlated by the model. The thickness of a line correlates with the number of associations found across different abstracts.

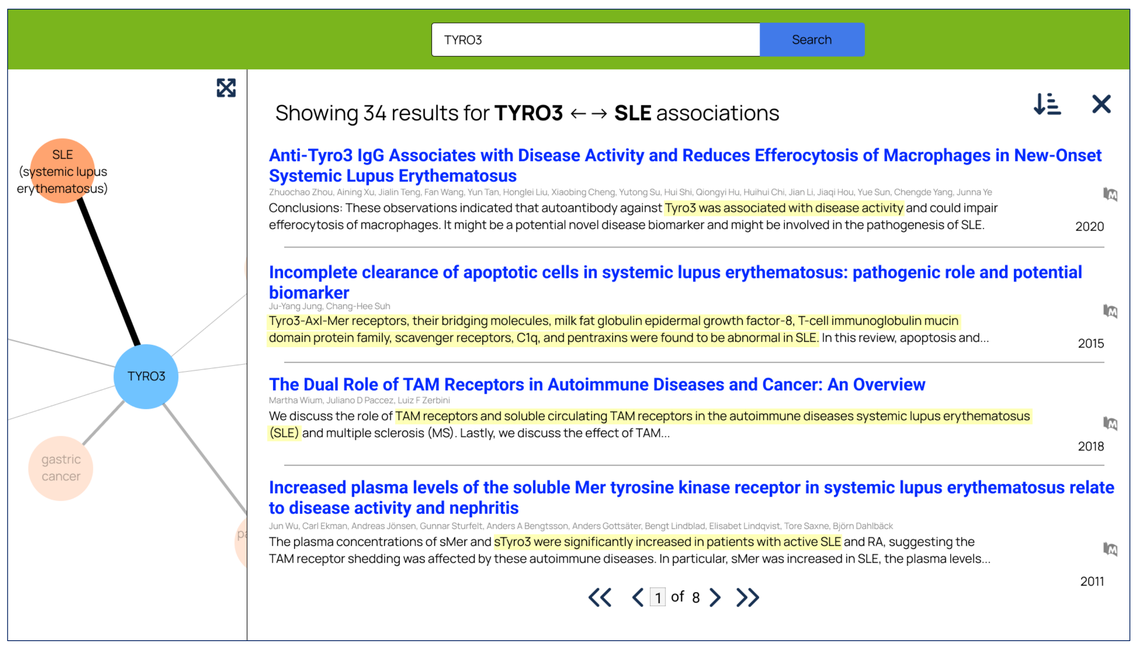

Fig. 3: Clicking on an edge or a node lists every article, where the correlation was determined by the algorithm. An excerpt of the abstract is shown below the title, with the relevant sentence highlighted. Clicking on the title leads to the article on PubMed.

Discussion and conclusion

Implementing new state-of-the-art Natural Language Processing architectures opens new possibilities for text mining and research discovery. Our tool is easy to use and not only limited to expert researchers and bioinformaticians, as it can help anyone to (re)-discover biomedical literature, find unknown associations and generate research hypothesis on entities of interest. Additionally, we manually curated a high-quality relation extraction dataset, which is suitable for all kinds of machine learning models.

References

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UKM Universitätsklinikum Münster
WWU MÜNSTER
medizinische fakultät Westfälische Wilhelms-Universität Münster